ISOMETRIC TIGHT TORI WITH DIFFERENT ENCLOSED VOLUMES

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Abstract. We construct two smooth embedded tight tori in Euclidean three-space whose induced metrics agree and whose enclosed volumes differ. The construction uses one closed asymptotic curve from the annulus of Banks, Ghomi, Qiu, and Raffaelli. We modify the ends of that annulus to infinite-order planar contact and complete it to a tight torus. For any tight torus with one closed asymptotic curve in negative curvature, we prove that a modification supported near the curve produces an infinitesimal bending with nonzero normal flux. Gauss-map injectivity forces an inflection, a local change of the curve arranges a derivative condition, and a choice of rulings makes the nearby asymptotic curves periodic. A bending on this ruled annulus is then extended by zero to the completed surface. Opposite multiples of the bending give the required pair. Each stage is stated as a separate existence or implication theorem with explicit hypotheses.

1. Definitions, main result, and the three implications

All manifolds, maps, and fields are smooth, meaning C^∞, unless another regularity is specified. A *closed surface* is a compact connected smooth two-dimensional manifold without boundary. We use the standard inner product, orientation, cross product, and Lebesgue volume in $\mathbb{R}^3$. Write $T^2 = (\mathbb{R}/2\pi\mathbb{Z})^2$. An *embedding* is an injective immersion that is a homeomorphism onto its image.

For an immersion $X : M \to \mathbb{R}^3$, its induced metric is

$$g_X(v, w) = \langle dX_p(v), dX_p(w)\rangle \quad (p \in M,\ v, w \in T_pM).$$

Write K_X and dA_X for the Gaussian curvature and area measure of this metric. If n is a unit normal, our second fundamental form has local coefficients

$$\mathrm{II}_{ij} = \langle X_{ij}, n\rangle.$$

The normal component makes this definition independent of the chosen coordinates. In a coordinate chart, if $g_{ij} = \langle X_i, X_j\rangle$, then

$$K_X = \frac{\mathrm{II}_{11}\mathrm{II}_{22} - \mathrm{II}_{12}^2}{g_{11}g_{22} - g_{12}^2}. \tag{1}$$

This is the Gauss equation; see [4, §3-3 and §4-3]. Reversing n changes II by a sign and leaves K_X unchanged.

Definition 1.1. An embedding $X : M \to \mathbb{R}^3$ of a closed surface is *tight* if, for every $e \in \mathbb{R}^3 \setminus \{0\}$ and $t \in \mathbb{R}$, the set

$$X(M) \cap \{x \in \mathbb{R}^3 : \langle x, e\rangle > t\}$$

is empty or connected. This is the two-piece property for surfaces.

By the Jordan–Brouwer separation theorem, the image of a closed embedded surface in $\mathbb{R}^3$ separates space into a bounded and an unbounded component. Its *enclosed volume* $\mathrm{Vol}(X)$ is the Lebesgue volume of the bounded component, and ν_X denotes its outward unit normal.

Two embeddings of the same source are *isometric by the identity* if their induced metrics are equal. Their images are *congruent* if a Euclidean isometry of $\mathbb{R}^3$ maps one image onto the other.

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Definition 1.2. A smooth map $Y : M \to \mathbb{R}^3$ is an *infinitesimal bending* of an immersion X if

$$\langle dX_p(v), dY_p(w)\rangle + \langle dX_p(w), dY_p(v)\rangle = 0 \quad (p \in M,\ v, w \in T_pM). \tag{2}$$

It is a vector field along X, and is not required to be tangent to $X(M)$. Its *support* is the closure in M of $\{p : Y(p) \neq 0\}$. For a closed embedded surface its *outward normal flux* is

$$\int_M \langle Y, \nu_X\rangle \, dA_X.$$

For subsets $A \subset U \subset M$, the notation $A \Subset U$ means that $\overline{A}$ is compact and contained in the open set U. The support of $X - X_0$ always refers to the pointwise difference of two maps on the same source.

Definition 1.3. A regular curve c on an immersed surface X is *asymptotic* if $\mathrm{II}(c', c') = 0$. Equivalently, for its image curve $\gamma = X \circ c$, the acceleration γ'' is tangent to the surface. For any change of regular parameter, the extra term in the acceleration is a multiple of γ', so this condition is independent of the parameter. When X is an embedding, a *closed embedded asymptotic curve* means the image of a smooth embedding $c : S^1 \to M$ with this property.

Theorem 1.4 (Main result)**.** *There exist a smooth tight embedding $X : T^2 \to \mathbb{R}^3$ and an infinitesimal bending $Y : T^2 \to \mathbb{R}^3$ such that*

$$\operatorname{supp} Y \Subset \{K_X < 0\}, \qquad \int_{T^2} \langle Y, \nu_X\rangle \, dA_X \neq 0.$$

There is $a_0 > 0$ such that for every $0 < a < a_0$ the maps $X_\pm = X \pm aY$ are smooth tight embeddings satisfying

$$g_{X_+} = g_{X_-}, \qquad \mathrm{Vol}(X_+) \neq \mathrm{Vol}(X_-). \tag{3}$$

Consequently their images are noncongruent. Both maps agree with X outside $\operatorname{supp} Y$*, and in particular on a nonempty open set.*

Here equality of metrics gives an intrinsic isometry by the identity map of T^2. We do not assert that the metrics of $X + tY$ are independent of t. The volume-stationarity problem for infinitesimal bendings is discussed by V. A. Aleksandrov [1].

The proof is organized through the following three statements. Their inputs and outputs are geometric objects on specified sources, so that the existence construction and the local deformation can be used independently.

Theorem 1.5 (Existence of a starting torus)**.** *There exist a smooth tight embedding $X_0 : T^2 \to \mathbb{R}^3$ and a smooth embedding $c : S^1 \to T^2$ such that $X_0 \circ c$ is asymptotic and $K_{X_0} < 0$ on $c(S^1)$.*

Theorem 1.5 is obtained in Section 3 from the construction of Banks–Ghomi–Qiu–Raffaelli [3]. Only one of their closed asymptotic curves is used. The completion includes planar regions, which are allowed in the present statement and remain fixed under the subsequent deformation.

Theorem 1.6 (One curve implies a supported bending)**.** *Suppose $X_0 : T^2 \to \mathbb{R}^3$ is a smooth tight embedding and $c : S^1 \to T^2$ is a smooth embedding such that $X_0 \circ c$ is asymptotic and $K_{X_0} < 0$ along $c(S^1)$. For every open set U with*

$$c(S^1) \subset U \subset \{K_{X_0} < 0\},$$

and every $\varepsilon > 0$, there exist a smooth tight embedding $X : T^2 \to \mathbb{R}^3$ and an infinitesimal bending Y of X such that

$$\begin{gathered} \operatorname{supp}(X - X_0) \Subset U, \qquad K_X < 0 \text{ on } U, \qquad \|X - X_0\|_{C^1} < \varepsilon, \\ \operatorname{supp} Y \Subset U, \qquad \int_{T^2} \langle Y, \nu_X\rangle \, dA_X \neq 0. \end{gathered} \tag{4}$$

The C^1 norm is computed using any fixed smooth Riemannian metric on the compact source and the Euclidean norm in the target. The same convention, with covariant derivatives for the chosen background metric, is used for C^2 norms below.

The bending in this theorem belongs to the modified embedding X. No condition on derivatives of the torsion, and no family of closed asymptotic curves, is required of $X_0 \circ c$. Sections 4–8 prove this theorem.

Proposition 1.7 (A supported bending implies an isometric pair)**.** *Suppose $X : M \to \mathbb{R}^3$ is a smooth tight embedding of a closed surface, and Y is an infinitesimal bending with*

$$\operatorname{supp} Y \Subset \{K_X < 0\}, \qquad \int_M \langle Y, \nu_X \rangle \, dA_X \neq 0.$$

Then, for all sufficiently small $a > 0$, $X_\pm = X \pm aY$ are tight embeddings with unequal enclosed volumes and

$$g_{X_\pm} = g_X + a^2 \langle dY, dY \rangle, \quad \langle dY, dY \rangle(v, w) = \langle dY(v), dY(w) \rangle. \tag{5}$$

Proposition 1.7 is proved in Section 2. Theorem 1.5 supplies exactly the input of Theorem 1.6; its output supplies exactly the input of Proposition 1.7. This separates the global existence problem from the local bending problem.

Proof of Theorem 1.4 from these statements. Choose X_0, c by Theorem 1.5. Compactness of $c(S^1)$ and continuity of K_{X_0} give an open neighborhood U with $\overline{U} \subset \{K_{X_0} < 0\}$. Apply Theorem 1.6 and then Proposition 1.7. The positive-curvature locus of a tight surface is nonempty by Theorem 2.1 below, and is unchanged by these modifications. It supplies the stated common open set. □

2. Classical reductions and the volume argument

Theorem 2.1 (Classical curvature criterion for tightness)**.** *For a smooth embedding $X : M \to \mathbb{R}^3$ of a closed surface, the two-piece property of Definition 1.1 is equivalent to*

$$\int_M \max(K_X, 0) \, dA_X = 4\pi. \tag{6}$$

We use this classical theorem as an external result; see Banchoff–Kühnel [2, Definition 1.1.1, p. 57]. By Gauss–Bonnet, its equivalent total-absolute-curvature formulation is $\int_M |K_X| \, dA_X = 8\pi - 2\pi\chi(M)$. This criterion is useful here because it reduces preservation of tightness to preservation of the positive-curvature part. It allows zero-curvature regions.

Lemma 2.2 (Preservation under a negative-curvature modification)**.** *Suppose $X_0 : M \to \mathbb{R}^3$ is a tight embedding of a closed surface, $U \subset \{K_{X_0} < 0\}$ is open, and $X : M \to \mathbb{R}^3$ is an embedding with $\operatorname{supp}(X - X_0) \Subset U$ and $K_X < 0$ on U. Then X is tight. In particular these conditions hold for every sufficiently small C^2 perturbation supported in a fixed compact subset of $\{K_{X_0} < 0\}$.*

Proof. On U both positive curvature densities vanish. Outside the compact support of $X - X_0$, the maps agree on a neighborhood and hence have the same metric and curvature. Thus their total positive curvatures are equal, and Theorem 2.1 applies. For the last assertion choose a compact neighborhood of the prescribed support contained in $\{K_{X_0} < 0\}$. On this neighborhood the original curvature has a strictly negative upper bound. Formula (1) and continuity in the first two derivatives preserve that bound under a sufficiently small C^2 change. Embeddings of a compact source form an open set in the C^1 topology [7, Chapter 2, §1, Theorem 1.4]. □

Lemma 2.3 (First variation of enclosed volume)**.** *For an embedding $X : M \to \mathbb{R}^3$ of a closed surface and a smooth field $Y : M \to \mathbb{R}^3$, the maps $X + tY$ are embeddings for sufficiently small $|t|$, and*

$$\left.\frac{d}{dt}\right|_{t=0} \operatorname{Vol}(X + tY) = \int_M \langle Y, \nu_X \rangle \, dA_X. \tag{7}$$

For small t, the enclosed volume is a polynomial in t of degree at most three.

Proof. The assertion about embeddings follows from the openness just cited. Orient M so that X has outward normal. On $\mathbb{R}^3$ define the two-form

$$\beta = \tfrac{1}{3}(x\,dy\wedge dz + y\,dz\wedge dx + z\,dx\wedge dy), \qquad d\beta = dx\wedge dy\wedge dz.$$

Set $X_t = X + tY$ and $\mathcal{V}(t) = \int_M X_t^*\beta$. The divergence theorem gives $\mathcal{V}(0) = \mathrm{Vol}(X) > 0$. For every embedded X_t it also gives $\mathcal{V}(t) = \mathrm{Vol}(X_t)$ or $-\mathrm{Vol}(X_t)$, according to orientation. Continuity and positivity at zero select the positive sign for all sufficiently small $|t|$. In positively oriented local coordinates,

$$X_t^*\beta = \tfrac{1}{3}\det(X + tY, X_u + tY_u, X_v + tY_v)\,du\wedge dv.$$

The integrand is a polynomial of degree at most three, proving the polynomial assertion globally.

Extend the field Y off $X(M)$ by a tubular projection and use Cartan's formula for the derivative of pullback at zero. The exact term has integral zero by Stokes' theorem on the closed source. The remaining term is

$$\mathcal{V}'(0) = \int_M X^*(\iota_Y d\beta) = \int_M \langle Y, \nu_X\rangle\, dA_X.$$

Here ι_Y denotes contraction with the extended velocity field; its restriction to the surface is independent of the extension. This proves (7). The classical ingredients are Cartan's formula, Stokes' theorem, and the divergence theorem; see [9, Theorem 14.35 and Chapter 16, including Theorem 16.32]. □

Proof of Proposition 1.7. For arbitrary tangent vectors v, w, expand

$$\langle d(X + tY)(v), d(X + tY)(w)\rangle.$$

The linear term vanishes by (2), leaving $g_X(v,w) + t^2\langle dY(v), dY(w)\rangle$. This proves the exact identity (5) at $t = \pm a$. By Lemma 2.3,

$$\mathrm{Vol}(X + aY) - \mathrm{Vol}(X - aY) = 2a\int_M \langle Y, \nu_X\rangle\, dA_X + O(a^3). \tag{8}$$

The coefficient of a is nonzero; hence the difference is nonzero for every sufficiently small $a > 0$.

Choose a compact neighborhood of $\operatorname{supp} Y$ inside $\{K_X < 0\}$. For fixed Y, $X + tY \to X$ in C^2, so its curvature stays negative on that neighborhood for small $|t|$. Outside it the embedding is unchanged. Lemma 2.2 proves tightness. Euclidean isometries preserve enclosed volume, so unequal volumes exclude congruence of the images. □

3. The BGQR annulus and completion to a tight torus

The purpose of this section is only to prove Theorem 1.5. The local bending theorem will use the resulting torus and its one retained curve, without any formula for that curve. For the planar boundary curves below, *strictly convex* means that the curve is an embedded circle bounding a convex disk and has positive signed curvature in its counterclockwise orientation.

Definition 3.1. Let $S \subset \mathbb{R}^2 \times [-h, h]$, $h > 0$, be an embedded compact annulus whose boundary curves bound smooth planar disks $D_\pm \subset \{z = \pm h\}$. An *inward projected collar* at the upper boundary means that horizontal projection maps a neighborhood of that boundary in S diffeomorphically onto a collar of ∂D_+ inside D_+. On this collar the annulus is a graph $z = h + b_+(x)$. Infinite-order planar contact means that every partial derivative of b_+, including its value, is zero on ∂D_+. At the lower boundary the analogous graph is $z = -h + b_-(x)$ with the same condition. Smoothness and derivatives at the boundary are understood in coordinates on a manifold with boundary.

Proposition 3.2 (An annulus implies a tight completion)**.** *Suppose an embedded smooth compact annulus S has all the properties in Definition 3.1, its interior lies in $-h < z < h$, and $K_S \le 0$. Then it is contained in a smooth embedded tight torus which agrees with S throughout its interior.*

Proof. Choose $R > 0$ such that the two disks and the horizontal projection of S lie in $|x| < R$. To supply a convex outer surface with planar caps, choose a smooth function $\phi : \mathbb{R} \to [0, \infty)$ satisfying

$$\phi = 0 \text{ on } (-\infty, 0], \qquad \phi' > 0,\ \phi'' > 0 \text{ on } (0, \infty), \qquad \phi(t) \longrightarrow \infty.$$

For example, for $t > 0$ set $\phi(t) = \int_0^t (t-u)e^{-1/u}\,du$ and extend by zero. Consider the body

$$\Omega = \{(x, z) \in \mathbb{R}^2 \times \mathbb{R} : \phi(|x|^2 - R^2) + z^2/h^2 \le 1\}. \tag{9}$$

Its defining function is convex, it has a minimum zero at the origin, and the sublevel set is compact with nonempty interior. On level one the gradient cannot vanish: if $z \neq 0$ its vertical component is nonzero, while if $z = 0$, then $|x| > R$ and its horizontal component is $2\phi' x \neq 0$. Thus $\partial\Omega$ is smooth. A compact convex body with nonempty interior has boundary homeomorphic to a sphere; here radial projection from the origin also gives a diffeomorphism, by transversality and the implicit function theorem.

The two planar caps are $|x| \le R$, $z = \pm h$. Away from them, $|x| > R$, and the defining Hessian is

$$\begin{pmatrix} 2\phi' I + 4\phi'' x \otimes x & 0 \\ 0 & 2/h^2 \end{pmatrix},$$

which is positive definite. Its restriction to the tangent plane, divided by the length of the gradient, is the second fundamental form up to sign. Thus $\partial\Omega$ has positive Gaussian curvature away from the caps and zero curvature on the caps. Gauss–Bonnet on this convex sphere gives total curvature 4π.

Every interior point of S lies strictly inside Ω: its horizontal term in (9) is zero and $|z| < h$. Remove the interiors of D_+ and D_- from $\partial\Omega$ and attach S along their boundary curves. In horizontal coordinates, the planar piece occupies the exterior side of each disk boundary, and the annulus occupies the interior side. The graph defining the union is $b_\pm$ on one side and zero on the other; the vanishing of all boundary derivatives proves that this graph is smooth. This also proves immersion along the seam. Strict interiority excludes intersections between the other points of S and $\partial\Omega$.

The union is a connected closed embedded surface. It is orientable, as every closed embedded surface in $\mathbb{R}^3$ is. Removing two disks from a sphere leaves an annulus, so its Euler characteristic after attaching S is zero. The classification of closed orientable surfaces therefore identifies it with a torus. The removed disks have zero curvature, the retained convex part contributes positive curvature 4π, and S contributes none. Its total positive curvature is consequently 4π. Theorem 2.1 proves tightness. □

Theorem 3.3 (External construction: BGQR)**.** *There exist a smooth embedded compact annulus S_0, negatively curved in its interior, and a regular embedded closed asymptotic curve Γ in that interior. The annulus has a parametrization*

$$S_0(\theta, r) = (C(\theta) - re_\theta, z(r)), \qquad (\theta, r) \in (\mathbb{R}/2\pi\mathbb{Z}) \times [-200, 200],$$

where $e_\theta = (\cos\theta, \sin\theta)$ and $J(x, y) = (-y, x)$. There are periodic functions P, Q such that

$$C' = Pe_\theta + QJe_\theta, \quad P > 0.$$

The function z is odd, $z(r) = r$ for $|r| \le 175$, $z' > 0$ for $|r| < 200$, $rz'' \le 0$, and

$$z(\pm 200) = \pm 575/3, \qquad z'(\pm 200) = 0, \qquad z''(\pm 200) = \mp 1/15.$$

Writing $v = Q - r$ and $D = P^2 + v^2 + PQ' - vP'$, the end regions $175 \le |r| \le 200$ satisfy $D > 0$ and $rv < 0$. The endpoint sections are strictly convex planar Jordan curves.

This is an external input: Theorem 1.1, Section 2.4, Proposition 3.2, and Proposition 3.3 of [3], specifically equation (7) in Section 3.1 and equations (11), (13), and (14) in Section 3.2. The theorem there supplies two curves; we select one. The numerical and coordinate assertions above are outputs of that particular construction, rather than consequences of its abstract Theorem 1.1 alone. We do not reproduce its construction. The source boundary contact is not infinite-order planar contact, so Proposition 3.2 still requires the following modification.

Lemma 3.4 (Flattening the ends while retaining the curve)**.** *The annulus in Theorem 3.3 can be modified outside a neighborhood of* Γ *so that it satisfies all the hypotheses of Proposition 3.2, with* $K < 0$ *in its interior.*

Proof. Put $h_0 = 575/3$. Since Γ is compactly contained in the interior and z is strictly increasing there, choose

$$\max\{175, \max_{\Gamma} |z|\} < h_1 < h_0.$$

Choose an even smooth function $q : \mathbb{R} \to [0, 1]$ which equals one on $[-h_1, h_1]$, is positive on $(-h_0, h_0)$, vanishes outside this interval, and is nonincreasing on $[0, \infty)$. To construct it, take the nonnegative bump $e^{-1/[t(1-t)]}$ on $(0, 1)$, extended by zero, and normalize its cumulative integral. Rescaling this smooth step function and subtracting from one gives the required transition on $[h_1, h_0]$; reflect evenly. In particular all its derivatives vanish at $\pm h_0$. Set

$$f(t) = \int_0^t q(a)\, da, \qquad H(r) = f(z(r)), \qquad h = f(h_0) > 0.$$

Then f is odd, equals the identity on $[-h_1, h_1]$, $f' > 0$ in $(-h_0, h_0)$, and $tf''(t) \le 0$. The new annulus is

$$S(\theta, r) = (C(\theta) - re_\theta, H(r)).$$

It agrees with S_0 in a neighborhood of Γ. On the interior, $H' > 0$; hence equality of two points of S first gives equality of r, and then injectivity follows from that of S_0. Its differential has rank two there. At an endpoint the horizontal derivatives have determinant $v \neq 0$, so immersion also holds at the boundary. The height separates the two endpoint sections from all other sections. Thus S is an embedding of the compact annulus.

For completeness the curvature calculation is given explicitly. In the orthonormal horizontal basis (e_θ, Je_θ), write $w = C - re_\theta$. Then

$$w_\theta = Pe_\theta + vJe_\theta, \quad w_r = -e_\theta, \quad w_{\theta\theta} = (P' - v)e_\theta + (P + Q')Je_\theta.$$

The area vector of S is

$$S_\theta \times S_r = (H'v, -H'P, v)$$

in the basis $(e_\theta, Je_\theta, e_z)$, and its squared length is $H'^2(P^2 + v^2) + v^2$. The unnormalized second coefficients are

$$\langle S_{\theta\theta}, S_\theta \times S_r\rangle = -H'D, \quad \langle S_{\theta r}, S_\theta \times S_r\rangle = H'P, \quad \langle S_{rr}, S_\theta \times S_r\rangle = H''v.$$

Consequently

$$K_S = -\frac{H'^2P^2 + H'H''vD}{[H'^2(P^2 + v^2) + v^2]^2}. \tag{10}$$

Where the height changes, $|z| > h_1 > 175$ implies $|r| > 175$. Since z has the sign of r,

$$rH'' = r\{f''(z)z'^2 + f'(z)z''\} \le 0.$$

The inequalities $rv < 0$ and $D > 0$ therefore give $H''vD \ge 0$. The first numerator term in (10) is strictly positive in the interior because $H' > 0$ and $P > 0$. Elsewhere the original surface is unchanged. Thus $K_S < 0$ throughout the interior and $K_S = 0$ on the boundary.

It remains to verify the attachment conditions, including the side of the projected collars. The horizontal endpoint curves have tangent $Pe_\theta + vJe_\theta$ and determinant $D > 0$, so their counterclockwise orientation has outward unit normal

$$\frac{ve_\theta - PJe_\theta}{\sqrt{P^2 + v^2}}.$$

At the upper end the continuation towards increasing r is $-e_\theta$ and its dot product with this outward normal is $-v/\sqrt{P^2 + v^2} > 0$. At the lower end the continuation towards decreasing r is e_θ and the corresponding dot product is $v/\sqrt{P^2 + v^2} > 0$. The interior projected collars therefore lie

inside the bounded disks. Their projection is a local diffeomorphism, since its Jacobian is $v \neq 0$ at the boundary. Compactness of each boundary and its embedding provide a collar on which this projection is a diffeomorphism.

All positive-order derivatives of f at $\pm h_0$ vanish. The chain rule shows that $H - H(\pm 200)$ has every derivative zero at the corresponding endpoint. After the smooth horizontal change of coordinates just established, the graph functions also have every boundary derivative zero. These are exactly the conditions of Definition 3.1. □

Proof of Theorem 1.5. Apply Lemma 3.4 to the annulus and chosen curve in Theorem 3.3, and then apply Proposition 3.2. The curve and a neighborhood of it were retained unchanged, so it is still embedded, asymptotic, and in $K < 0$ on the completed tight torus. Parametrize that torus by T^2; the inverse parametrization of the retained curve gives the required embedding c. □

4. The Darboux frame and the forced inflection

The remaining sections prove Theorem 1.6 for any starting torus satisfying its hypotheses. The construction of Section 3 will not be used again.

Let $\gamma : \mathbb{R}/\ell\mathbb{Z} \to \mathbb{R}^3$, $\ell > 0$, be the given curve parametrized by arclength, and let $n(s)$ be the unit normal of the surface along it. Put $T = \gamma_s$ and $E = n \times T$. Then (T, E, n) is a positively oriented orthonormal frame. All derivatives in s are derivatives of periodic lifts.

Lemma 4.1 (Frame equations)**.** *There are smooth real periodic functions k, τ satisfying*

$$T_s = kE, \qquad E_s = -kT + \tau n, \qquad n_s = -\tau E. \tag{11}$$

They are given by $k = \langle T_s, E\rangle$ and $\tau = \langle E_s, n\rangle$. Moreover $K = -\tau^2$ along γ.

Proof. Unit speed gives $T_s \perp T$, and asymptoticity gives $T_s \perp n$, proving the first equation. Differentiating orthonormality gives the other two, with the T component of n_s equal to $-\langle n, T_s\rangle = 0$. In the orthonormal tangent basis (T, E), the second fundamental form has first entry zero and mixed entry τ. Its determinant is therefore $-\tau^2$, which proves the curvature identity by (1). □

Here k is the signed geodesic curvature and τ is the signed geodesic torsion. Negative Gaussian curvature implies $\tau \neq 0$; hence its sign is constant on the circle. Define

$$\rho = |\tau|^{-1/2} > 0. \tag{12}$$

This smooth periodic function equals the negative Gaussian curvature along the core raised to the power $-1/4$. We use this surface frame even at points where the curve's curvature vanishes, where a Frenet normal need not be defined. The exponent is chosen so that $\rho_s/\rho = -\tau_s/(2\tau)$, the integrating factor relation in the asymptotic return equation of Section 6.

Definition 4.2. An *inflection* of the unit-speed curve γ is a point s_0 where $\gamma_{ss}(s_0) = 0$. By (11) this is equivalent to $k(s_0) = 0$. No prescribed order of vanishing is included in this definition.

Lemma 4.3 (Injectivity of the negative Gauss map)**.** *For a smooth embedded tight torus with outward normal n, the restriction of n to $\{K < 0\}$ is injective.*

Proof. For a regular value $u \in S^2$ of n, let $m_+(u)$ and $m_-(u)$ be the numbers of preimages with positive and negative curvature. They are finite because the source is compact and each preimage is isolated. A maximum of the height function $\langle X, u\rangle$ has outward normal u. At that point its Hessian is negative semidefinite; regularity of u makes the Hessian nondegenerate, so it is negative definite and the Gaussian curvature is positive. Thus $m_+(u) \geq 1$ for every regular value.

The change-of-variables formula for the Gauss map, whose absolute Jacobian is $|K|$, and Theorem 2.1 give

$$\int_{S^2} m_+(u)\, dA(u) = \int_{\{K>0\}} K\, dA = 4\pi.$$

Sard's theorem makes the set of nonregular values have measure zero. Since S^2 has area 4π, it follows that $m_+ = 1$ almost everywhere. The signed-degree formula gives

$$m_+(u) - m_-(u) = \deg n = \frac{1}{4\pi}\int_{T^2} K\, dA = \frac{\chi(T^2)}{2} = 0.$$

Here the equality for the degree follows either from $n^*(dA_{S^2}) = K\, dA$ or the usual integration formula for degree, and the next equality is Gauss–Bonnet. Thus $m_- = 1$ almost everywhere as well. Change of variables, Sard's theorem, and the degree formula are used in their ordinary smooth-manifold forms; see [9, Chapters 6 and 17].

If two different points in $\{K < 0\}$ had the same normal, the inverse function theorem would provide disjoint neighborhoods mapped diffeomorphically onto overlapping open subsets of S^2. Every regular value in their nonempty open overlap would have at least two negative-curvature preimages. This contradicts $m_- = 1$ almost everywhere, since a nonempty open subset of S^2 has positive area. □

Proposition 4.4 (One loop supplies an inflection)**.** *Every regular embedded closed asymptotic curve in $\{K < 0\}$ on a smooth embedded tight torus has an inflection. In fact its signed geodesic curvature k takes both positive and negative values.*

Proof. Lemma 4.3 shows that $s \mapsto n(s)$ is injective. It is regular because $|n_s| = |\tau| > 0$. It is therefore a simple regular closed curve on the unit sphere.

We use the Gauss–Bonnet–Fenchel argument recorded by Ghomi–Raffaelli [6, Note 5.2, equations (10)–(12)], where it is attributed to Kovaleva [8]. We give the calculation to specify its orientations and its application here. Let σ be arclength on the spherical normal curve, and let κ_{sph} be its signed geodesic curvature for the outward orientation of S^2. The frame equations give

$$n_{ss} = -\tau_s E + \tau k T - \tau^2 n, \qquad n \times n_s = \tau T.$$

Consequently

$$\kappa_{\mathrm{sph}} = \frac{\langle n_{ss}, n \times n_s\rangle}{|n_s|^3} = \frac{k}{|\tau|}, \qquad d\sigma = |\tau|\, ds,$$

and therefore $\int \kappa_{\mathrm{sph}}\, d\sigma = \int_0^\ell k\, ds$.

Choose the disk on the sphere lying to the left of the oriented normal curve, and let its area be A. The Jordan curve theorem on the sphere gives two disks with nonempty open interiors, so $0 < A < 4\pi$. The sphere has curvature one and this disk has Euler characteristic one. Gauss–Bonnet with boundary gives

$$\int \kappa_{\mathrm{sph}}\, d\sigma + A = 2\pi, \qquad \left|\int_0^\ell k\, ds\right| = |2\pi - A| < 2\pi.$$

Gauss–Bonnet with boundary is used in its standard smooth-boundary form [4, §4-5]. No convexity or hemisphere condition on the spherical disk is required.

On the other hand Fenchel's theorem [5] for a regular closed space curve gives

$$\int_0^\ell |\gamma_{ss}|\, ds = \int_0^\ell |k|\, ds \geq 2\pi.$$

If k had only one sign, including the possibility of zeros, the last integral would equal $|\int k\, ds|$, a contradiction. Thus k has both signs, and continuity gives a zero. □

The role of the inflection is to permit a correction concentrated near a point where $k = 0$ without significantly changing the transverse component of a ruling. Before choosing that ruling, we arrange one further derivative condition. This is a modification of the starting geometry, rather than an additional hypothesis on the curve supplied by Theorem 1.5.

5. Local preparation of the curve

Proposition 5.1 (An inflection implies a prepared curve)**.** *Let $X_0 : M \to \mathbb{R}^3$ be a smooth embedding of a closed oriented surface, and let $c : \mathbb{R}/\ell\mathbb{Z} \to M$ be a smooth embedding such that $\gamma = X_0 \circ c$ is a unit-speed asymptotic curve in $\{K_{X_0} < 0\}$. Use its frame from Section 4, and suppose $k(s_0) = 0$. For every open neighborhood U of $c(\mathbb{R}/\ell\mathbb{Z})$ with $U \subset \{K_{X_0} < 0\}$ and every $\varepsilon > 0$, there is a smooth embedding $\widetilde{X} : M \to \mathbb{R}^3$ such that*

$$\operatorname{supp}(\widetilde{X} - X_0) \Subset U, \quad \|\widetilde{X} - X_0\|_{C^2} < \varepsilon, \quad K_{\widetilde{X}} < 0 \text{ on } U.$$

The actual curve $\widetilde{X} \circ c$ is closed, embedded, and asymptotic. In its own arclength σ, chosen to be zero at $c(s_0)$, its frame functions satisfy

$$\widetilde{k}(0) = 0, \qquad \widetilde{\rho}_{\sigma\sigma}(0) \neq \widetilde{\rho}(0)^{-3}. \tag{13}$$

Here $\widetilde{\rho} = |\widetilde{\tau}|^{-1/2}$ for the new curve. If X_0 is tight, $\widetilde{X}$ is tight.

Proof. We change the actual curve while retaining its tangent directions and its normal directions. Prescribe a new velocity $v(s)T(s)$ with $v > 0$. To keep the curve closed it is necessary and sufficient that

$$\int_0^\ell (v-1)T\,ds = 0, \tag{14}$$

because $\int_0^\ell T\,ds = 0$. We construct a family $v = v_\delta$, $\delta \downarrow 0$, with

$$v(s_0) = 1, \quad v_s(s_0) = 0, \quad v_{ss}(s_0) = d, \qquad \|v-1\|_{C^1} \longrightarrow 0, \tag{15}$$

where the fixed real number d will be chosen below.

There is an interval disjoint from s_0 on which $k \neq 0$. Indeed $k \equiv 0$ would make T constant, contradicting $\int T\,ds = 0$ and $|T| = 1$; a nonzero value has an open neighborhood away from the zero at s_0. On such an interval,

$$T_s = kE, \quad T_{ss} = -k^2T + k_sE + k\tau n, \quad \det(T, T_s, T_{ss}) = k^2\tau \neq 0.$$

It follows that the vectors $T(s)$ on that interval span $\mathbb{R}^3$: otherwise they and their first two derivatives would lie in a fixed proper linear subspace, contradicting the determinant. Choose three points with linearly independent tangent vectors, and choose smooth bumps b_1, b_2, b_3 of integral one in disjoint sufficiently small neighborhoods of those points. The three vectors $\int b_iT\,ds$ are then linearly independent.

Choose a smooth cutoff χ on $\mathbb{R}$, equal to one near zero and supported in $(-1,1)$, and set in a coordinate interval at s_0

$$h_\delta(s) = \tfrac{d}{2}(s-s_0)^2\chi((s-s_0)/\delta),$$

extended by zero periodically. Its value and first derivative at s_0 are zero, and its second derivative there is d. Moreover $\|h_\delta\|_{C^1} = O(\delta)$ and $|\int h_\delta T\,ds| = O(\delta^3)$. Solve the invertible three-dimensional linear system

$$\sum_{i=1}^3 a_i \int b_iT\,ds = -\int h_\delta T\,ds.$$

Its matrix is fixed, so $a_i = O(\delta^3)$. Then $v = 1 + h_\delta + \sum_i a_ib_i$ satisfies (14) and (15). For small δ it is positive.

Integrate $\widetilde{\gamma}_s = vT$ with $\widetilde{\gamma}(s_0) = \gamma(s_0)$. The moment condition proves periodicity; estimates on $v-1$ and v_s show $\widetilde{\gamma} - \gamma \to 0$ in C^2. Since the surface is oriented, its embedded source circle has a two-sided smooth collar. Choose coordinates (s,w) in such a collar, with $X_0(s,0) = \gamma(s)$, whose closure lies in U. For a fixed cutoff $\chi_0(w)$ equal to one near zero and supported in the collar, define

$$\widetilde{X}(s,w) = X_0(s,w) + \chi_0(w)(\widetilde{\gamma}(s) - \gamma(s)),$$

and put $\widetilde{X} = X_0$ elsewhere. This tends to X_0 in C^2. Embedding and negative curvature persist, and Lemma 2.2 preserves tightness when present.

At $w = 0$, the two tangent vectors of $\widetilde{X}$ are vT and the original transverse tangent. Their span and orientation are unchanged; hence the normal is exactly $n(s)$. The new acceleration $\widetilde{\gamma}_{ss} = v_s T + vkE$ is tangent to this plane, so the new curve is asymptotic. Let σ be its actual arclength, with $\sigma(s_0) = 0$. Its new period is $\widetilde{\ell} = \int_0^\ell v(s)\, ds$; vector closure does not require this number to equal ℓ. Then

$$d\sigma = v\, ds, \qquad \widetilde{k} = k/v, \qquad \widetilde{\tau} = \tau/v, \qquad \widetilde{\rho} = \sqrt{v}\, \rho.$$

At s_0, where $v = 1$ and $v_s = 0$, differentiating twice yields

$$\widetilde{k} = 0, \qquad \widetilde{\rho}_{\sigma\sigma} = \rho_{ss} + \tfrac{1}{2}\rho d, \qquad \widetilde{\rho}^{-3} - \widetilde{\rho}_{\sigma\sigma} = \rho^{-3} - \rho_{ss} - \tfrac{1}{2}\rho d.$$

Choose d different from $2(\rho^{-3} - \rho_{ss})(s_0)/\rho(s_0)$. The last expression is then nonzero, proving (13) in the new arclength. A change of parameter alone would not change this geometric condition; the construction has changed the curve and the containing surface. □

6. From a prepared curve to a periodic ruled annulus

The next statement only uses a framed space curve. It is independent of any completion to a closed surface. Its output will be inserted into the prepared surface in Section 8.

Proposition 6.1 (Prepared curve implies periodic ruled annuli)**.** *Suppose $\gamma : \mathbb{R}/\ell\mathbb{Z} \to \mathbb{R}^3$ is a smooth embedded unit-speed curve with a smooth positively oriented orthonormal frame (T, E, n) satisfying $T = \gamma_s$, (11), and $\tau \neq 0$. Put $\rho = |\tau|^{-1/2}$. If for some s_0*

$$k(s_0) = 0, \qquad \rho(s_0)^{-3} - \rho_{ss}(s_0) \neq 0,$$

then there exist smooth periodic functions $f > 0$, g, and ω such that, for $V = gT + fE$:

1. *$R(s, u) = \gamma(s) + uV(s)$ is an embedded negatively curved ruled strip on $|u| < a$ for some $a > 0$, with the prescribed tangent plane $\operatorname{span}(T, E)$ along $u = 0$;*
2. *$\det(\gamma_s, V, V_s) = f^2\tau \neq 0$;*
3. *for all sufficiently large c,*

$$B(s, c) = \gamma(s) + \frac{\rho(s)}{c + \omega(s)} V(s)$$

has closed asymptotic constant-c curves. For every bounded nonempty open interval of such c, this is an embedding of the corresponding open annulus, and its image approaches the core as the interval is moved towards $+\infty$.

Proof. Write a transverse vector in the prescribed tangent plane as $V = gT + fE$, where $f > 0$. Its scale is part of the choice: fixing that scale in advance would impose an unnecessary nonlinear condition on the return equation. We impose the linear relation

$$f_s = -kg. \tag{16}$$

For the moment f, g are unspecified. Define the geometric component $\xi = \langle V_s, T\rangle = g_s - kf$. Then

$$V_s = \xi T + f\tau n, \qquad V_{ss} = \xi_s T + (k\xi - f\tau^2)E + (f_s\tau + f\tau_s)n.$$

In particular

$$\Delta := \det(T, V, V_s) = f^2\tau \neq 0. \tag{17}$$

At $u = 0$ the ruling is transverse to the core, so R is an immersion near it. Its mixed unnormalized second coefficient is $\langle R_{su}, R_s \times R_u\rangle = \Delta$ and $R_{uu} = 0$. Formula (1) therefore gives

$$K_R = -\frac{\Delta^2}{|R_s \times R_u|^4} < 0 \tag{18}$$

where R is immersed.

We determine the conditions for the nonruling asymptotic curves to close. Their graphs are $s \mapsto R(s, u(s))$. The acceleration has normal component equal to that of $R_{ss} + 2u_s R_{su}$: the $u_{ss} R_u$ term is tangent and $R_{uu} = 0$. Set

$$2A = \xi_s - \frac{\tau_s}{\tau}\xi + g\tau^2.$$

The frame equations give the two unnormalized coefficients

$$\langle R_{ss}, R_s \times V\rangle = f^2\tau_s u - 2f^2\tau A u^2, \qquad \langle R_{su}, R_s \times V\rangle = f^2\tau.$$

Thus the asymptotic equation is the Riccati equation

$$u_s = \frac{\rho_s}{\rho}u + Au^2. \tag{19}$$

Here $\tau_s/\tau = -2\rho_s/\rho$ follows from $\rho = |\tau|^{-1/2}$. For $u \neq 0$, taking the reciprocal gives

$$(\rho/u)_s = -\rho A. \tag{20}$$

It follows that periodicity is equivalent to the single condition $\int_0^\ell \rho A\, ds = 0$. Indeed integration over a period proves necessity; if this integral vanishes, a periodic function ω with $\omega_s = -\rho A$ exists, and every nonzero solution with positive denominator is

$$u(s, c) = \frac{\rho(s)}{c + \omega(s)}. \tag{21}$$

The parameter c is the additive constant in ρ/u; adding a constant to ω only changes this label.

It remains to choose f, g so that both (16) and the period condition hold. The point of retaining the scale f is that these requirements become linear. Using $\tau^2 = \rho^{-4}$ and $g_s = \xi + kf$, direct differentiation gives

$$2\rho A - \{g(\rho^{-3} - \rho_{ss}) - fk\rho_s\} = (\rho\xi + \rho_s g)_s. \tag{22}$$

The derivative on the right has integral zero. Consequently it suffices to solve

$$f_s + kg = 0, \qquad \int_0^\ell \{g(\rho^{-3} - \rho_{ss}) - fk\rho_s\}\, ds = 0, \qquad f > 0. \tag{23}$$

We now use the prepared inflection. Since a closed unit-speed curve cannot have constant tangent, $\int_0^\ell k^2\, ds > 0$. Let φ_δ be a nonnegative smooth function of integral one supported in the δ-neighborhood of s_0 on the circle. Define

$$b_\delta = \varphi_\delta - \frac{\int_0^\ell k\varphi_\delta\, ds}{\int_0^\ell k^2\, ds}\, k, \qquad (P_\delta)_s = kb_\delta, \qquad P_\delta(s_*) = 0,$$

where s_* is any fixed base point. The integral of kb_δ is zero, so P_δ is periodic. Since $k(s_0) = 0$,

$$\int_0^\ell |kb_\delta|\, ds \longrightarrow 0, \quad \|P_\delta\|_\infty \longrightarrow 0, \quad \int_0^\ell b_\delta F\, ds \longrightarrow F(s_0)$$

for every fixed continuous periodic F. The first assertion follows by bounding the bump contribution by $\sup_{|s-s_0|<\delta} |k(s)|$ and the correction by a constant times $|\int k\varphi_\delta|$; both tend to zero. The second follows by integration, and the third follows from concentration of φ_δ and the vanishing correction.

Set

$$\lambda_\delta = -\frac{\int_0^\ell k\rho_s\, ds}{\int_0^\ell \{b_\delta(\rho^{-3} - \rho_{ss}) + P_\delta k\rho_s\}\, ds}, \qquad f = 1 + \lambda_\delta P_\delta, \quad g = -\lambda_\delta b_\delta. \tag{24}$$

The denominator converges to $\rho(s_0)^{-3} - \rho_{ss}(s_0) \neq 0$. Thus λ_δ is bounded for small δ, and $f \to 1$ uniformly, giving $f > 0$. The first equality in (23) follows by differentiating f. The second follows by substitution of (24). Fix one such δ once and for all. This fixes the smooth ruling V, all its derivatives, and a periodic ω.

For completeness, compactness first gives a uniform width on which R is immersed. It is a local embedding at every $(s,0)$ by transversality. If no narrower strip were injective, there would be pairs of distinct parameters with $|u|, |u'| \to 0$ and equal images. Subsequences of their circle coordinates converge. Equality of the limiting core points and embeddedness of γ force both limits to be the same s. A local embedding near $(s,0)$ then excludes those pairs, a contradiction. After narrowing once more, injectivity on the compact closed cylinder gives a homeomorphism onto its image. Its restriction to the interior is therefore an embedded strip of some width $a > 0$.

Finally, for c sufficiently large, $c + \omega > 0$ and $0 < u(s,c) < a$, uniformly in s. The map $(s,c) \mapsto (s, u(s,c))$ is a diffeomorphism onto its image because

$$u_c = -\frac{\rho}{(c+\omega)^2} < 0$$

and its inverse is $c = \rho(s)/u - \omega(s)$. Composing with R proves the asserted annular embedding. Equation (20) proves that its constant-c circles are asymptotic. Their distance from the core tends uniformly to zero as $c \to +\infty$. The width was chosen after the ruling; no bound uniform in δ on derivatives of V is required. □

7. A reciprocal annulus implies a supported bending

This section gives a separate local result whose input is the annulus itself. In particular the result does not require the core to be parametrized by arclength or the function ρ to have any prescribed curvature interpretation. In its application they will be exactly the objects supplied by Proposition 6.1.

Theorem 7.1. *Let $\ell > 0$, and let $\gamma, V : \mathbb{R}/\ell\mathbb{Z} \to \mathbb{R}^3$ and $\rho, \omega : \mathbb{R}/\ell\mathbb{Z} \to \mathbb{R}$ be smooth periodic maps with $\rho > 0$ and*

$$\Delta(s) = \det(\gamma_s, V, V_s) \neq 0.$$

Let I be a nonempty open interval contained in $(-\min\omega, \infty)$. Suppose

$$B(s,c) = \gamma(s) + u(s,c)V(s), \qquad u(s,c) = \frac{\rho(s)}{c+\omega(s)} \tag{25}$$

is an embedding on $(\mathbb{R}/\ell\mathbb{Z}) \times I$, and each constant-$c$ circle is asymptotic on this annulus. Then B has a smooth infinitesimal bending Y whose support is compactly contained in this parameter annulus and whose signed normal flux satisfies

$$\int_{(\mathbb{R}/\ell\mathbb{Z})\times I} \det(Y, B_s, B_c)\, ds\, dc \neq 0. \tag{26}$$

The two coordinates are asymptotic: the constant-c curves are asymptotic by hypothesis, and the constant-s curves lie on straight rulings, even though c is not an affine parameter on them. We first solve the strain equations along these closed curves and then choose the remaining one-variable function to avoid cancellation of the flux.

7.1. Solving all three strain equations.

Construction of the bending. Let $F \in C_c^\infty(I)$, to be chosen later. The notation $C_c^\infty(I)$ means the smooth real functions on I whose support is a compact subset of I. Define

$$Q = \sqrt{|\Delta|}\, u_c, \qquad p = QF(c), \qquad Y \cdot V = 0, \quad Y \cdot B_s = p, \quad Y \cdot V_s = \frac{p_c}{2u_c}. \tag{27}$$

Here $u_c < 0$, so Q never vanishes. Also $B_s = \gamma_s + u_s V + uV_s$, whence $\det(B_s, V, V_s) = \Delta$. The three pairing equations therefore determine exactly one smooth field Y. The data are periodic in s, so Y is periodic as well. Its support is contained in $(\mathbb{R}/\ell\mathbb{Z}) \times \operatorname{supp} F$ and hence is compact in the open annulus. Derivatives of F do not enlarge its support.

We verify (2) on the coordinate pairs cc, sc, ss. Since $B_c = u_c V$ and V is independent of c, differentiation of $Y \cdot V = 0$ gives $Y_c \cdot B_c = 0$. For the mixed equation, put $z = p_c/(2u_c)$ temporarily. Differentiating the pairings gives

$$Y_s \cdot V = -z, \qquad Y_c \cdot B_s = p_c - Y \cdot B_{sc} = p_c - u_c z,$$

because $B_{sc} = u_{cs} V + u_c V_s$. Thus $Y_s \cdot B_c + Y_c \cdot B_s = -u_c z + p_c - u_c z = 0$.

For the ss equation, asymptoticity means that B_{ss} is tangent. As B_s, V are independent, write $B_{ss} = \alpha B_s + \beta V$ with smooth coefficients. Differentiating $\det(B_{ss}, B_s, V) = 0$ in c yields

$$2u_{cs}\Delta + u_c \det(B_s, V, V_{ss}) - \alpha u_c \Delta = 0.$$

Differentiating $\Delta = \det(B_s, V, V_s)$ in s yields

$$\Delta_s = \alpha \Delta + \det(B_s, V, V_{ss}).$$

Eliminating the last determinant proves

$$\alpha = \frac{\Delta_s}{2\Delta} + \frac{u_{cs}}{u_c} = \frac{Q_s}{Q}. \tag{28}$$

Since F is constant in s, $p_s = \alpha p$. It follows that $Y_s \cdot B_s = p_s - Y \cdot B_{ss} = p_s - \alpha p = 0$. All three components of the symmetric strain vanish. □

Remark 7.2 (The Codazzi interpretation of the factor)**.** The factor Q in (27) has a geometric meaning. Let $\mathcal{W} = |B_s \times B_c|$ be the area density and let $\sigma = \mathrm{II}_{sc}$ for the parameter unit normal. Then

$$|\sigma|\mathcal{W} = |\det(B_{sc}, B_s, B_c)| = |\Delta| u_c^2 = Q^2.$$

In asymptotic coordinates, the Mainardi–Codazzi equations give $\partial_s \log|\sigma| = \Gamma^s_{ss} - \Gamma^c_{sc}$, while the area-density identity gives $\partial_s \log \mathcal{W} = \Gamma^s_{ss} + \Gamma^c_{sc}$. Here the Γ's are the Christoffel coefficients of the induced metric in (s, c) coordinates. Adding the two equalities shows $\Gamma^s_{ss} = \frac{1}{2}\partial_s \log(|\sigma|\mathcal{W})$. Since $B_{ss} = \alpha B_s + \beta V$, its coefficient Γ^s_{ss} is α. This recovers (28) from the classical theory [4, §4-3]. The preceding determinant computation proves the precise identity needed without requiring any separate bending theorem.

7.2. **Computing and selecting the flux.**

Completion of the proof of Theorem 7.1. We must show that some choice of F in (27) has nonzero flux. Introduce the area vector for the affine ruling parameter and its squared length:

$$N(s,u) = (\gamma_s + uV_s) \times V, \qquad S(s,u) = |N(s,u)|^2.$$

Then $B_s \times B_c = u_c N$. Solving the pairing equations gives

$$Y = \frac{p(V \times V_s) + (p_c/(2u_c))N}{\Delta}.$$

Indeed the three scalar products are those of (27); uniqueness therefore verifies the formula. Since $N_u = V_s \times V$, one has $(V \times V_s) \cdot N = -S_u/2$. Consequently

$$\det(Y, B_s, B_c) = \frac{S p_c - u_c S_u p}{2\Delta}.$$

As Δ is independent of c and $S_c = u_c S_u$, subtracting $\partial_c(Sp/(2\Delta))$ leaves $-u_c S_u p/\Delta$. Compact support of F removes this total derivative under integration. Thus

$$\int \det(Y, B_s, B_c)\, ds\, dc = -\operatorname{sign}(\Delta) \int_I F(c) H(c)\, dc, \qquad H(c) = \int_0^\ell \frac{u_c^2 S_u}{\sqrt{|\Delta|}}\, ds. \tag{29}$$

The sign of Δ is constant since it is nonzero on a connected circle.

The function N is affine in u, so write $S = S_0 + S_1 u + S_2 u^2$, where the coefficients are smooth periodic functions of s and $S_2 = |V_s \times V|^2 > 0$ by $\Delta \neq 0$. Substituting u and u_c gives

$$H(c) = \int_0^\ell \frac{1}{\sqrt{|\Delta|}} \left\{ \frac{\rho^2 S_1}{(c+\omega)^4} + \frac{2\rho^3 S_2}{(c+\omega)^5} \right\} ds. \tag{30}$$

Although the annulus need only be embedded for $c \in I$, this scalar expression is defined on the whole connected interval $c > -\min \omega$.

It is real analytic on that interval. To see this, fix c_0 there and let $a = \min_s(c_0 + \omega(s)) > 0$. For $|c - c_0| < a/2$, the binomial power series for both denominator powers in (30) converge absolutely and uniformly in s, with a summable bound on every smaller parameter interval. Termwise integration gives a convergent power series for H. Only analyticity in the parameter c is used; the geometric data need only be smooth.

We next prove that H is not identically zero. Set $\epsilon = c + \min \omega > 0$ and $r(s) = \omega(s) - \min \omega \geq 0$. The integrand in (30) can be written

$$\frac{a(s)(\epsilon + r(s)) + b(s)}{(\epsilon + r(s))^5}, \qquad a(s) = \frac{\rho^2 S_1}{\sqrt{|\Delta|}}, \quad b(s) = \frac{2\rho^3 S_2}{\sqrt{|\Delta|}}.$$

These functions are bounded, and $m = \min_s b(s) > 0$. Let $M = \max_s |a(s)|$. Choose $r_0 > 0$ with $2Mr_0 \leq m/2$, with any $r_0 > 0$ if $M = 0$. For $0 < \epsilon \leq r_0$ and $r(s) \leq r_0$, the numerator is at least $m/2$. On $\{r \geq r_0\}$ the integrand is uniformly bounded in absolute value as $\epsilon \downarrow 0$.

Choose any minimum point s_*. Smoothness gives $r_s(s_*) = 0$ and $r(s_* + t) \leq Ct^2$ in a coordinate interval, for a fixed $C > 0$. For small ϵ, the interval $|t| \leq \sqrt{\epsilon}$ lies in $\{r \leq r_0\}$ and $\epsilon + r(s_* + t) \leq (1 + C)\epsilon$. Its integral is therefore bounded below by a positive constant times $\epsilon^{-9/2}$. All the rest of $\{r \leq r_0\}$ contributes nonnegatively, and the remaining region contributes a uniformly bounded quantity. We have proved

$$H(c) \longrightarrow +\infty \quad \text{as } c \downarrow -\min \omega.$$

This argument applies to degenerate, nonisolated, or multiple minima.

By the identity theorem for real analytic functions, H cannot vanish identically on any nonempty open subinterval of its domain. Choose a nonzero nonnegative $\chi \in C_c^\infty(I)$ and set $F = \chi H$. Then $F \in C_c^\infty(I)$ and

$$\int_I F(c) H(c)\, dc = \int_I \chi(c) H(c)^2\, dc > 0.$$

Indeed $\chi > 0$ on some open interval, on which H cannot vanish identically. Equation (29) now proves (26). The extension of the scalar formula outside I was used only to prove this nonidentity. The bending itself is defined and supported entirely on the actual embedded annulus. □

8. Insertion and passage to a global bending

The ruled annulus just constructed has the correct tangent planes along its core, but need not already be part of the original surface. The next proposition places that same ruled strip in the surface. Its curvature estimate is needed because cutting off a quadratic normal displacement need not be small in C^2.

Proposition 8.1 (A tangent ruled strip can be inserted)**.** *Let $X_0 : M \to \mathbb{R}^3$ be a smooth embedding of a closed oriented surface. Let $c : \mathbb{R}/\ell\mathbb{Z} \to M$ be an embedded asymptotic curve with image $\gamma = X_0 \circ c$, parametrized by arclength, and let U be an open source neighborhood contained in $\{K_{X_0} < 0\}$. Suppose $R(s,u) = \gamma(s) + uV(s)$ is an embedded strip near $u = 0$, with $V(s)$ tangent to $X_0(M)$ and transverse to γ_s. For every $\varepsilon > 0$ there is a smooth embedding $\widetilde{X} : M \to \mathbb{R}^3$ such that*

$$\operatorname{supp}(\widetilde{X} - X_0) \Subset U, \qquad \|\widetilde{X} - X_0\|_{C^1} < \varepsilon, \qquad K_{\widetilde{X}} < 0 \text{ on } U,$$

and there exist $a > 0$ and a smooth embedding $q : (\mathbb{R}/\ell\mathbb{Z}) \times (-a, a) \to U$ with

$$q(s, 0) = c(s), \qquad \widetilde{X}(q(s, u)) = R(s, u).$$

If X_0 is tight, $\widetilde{X}$ is tight.

Proof. Use an ambient tubular neighborhood of $X_0(M)$ with its normal projection π onto $X_0(M)$; see [9, Theorem 6.24]. After narrowing the strip, define

$$q = X_0^{-1} \circ \pi \circ R.$$

At $u = 0$, $q(s, 0) = c(s)$, and dq is invertible because R_s, R_u span the original tangent plane. Thus q is a local diffeomorphism near the core. It is a diffeomorphism onto a source neighborhood for a sufficiently narrow uniform strip. The proof of uniform injectivity is the same compactness argument as in Proposition 6.1, now applied to q and its embedded core. By the definition of normal projection, in that neighborhood

$$R \circ q^{-1} = X_0 + hn$$

for a smooth scalar function h and the original surface normal n. Equality of points and tangent planes along the core gives $h = dh = 0$ there.

Fix an original-source collar (s, w) of the core with compact closure in U. Narrow it to lie in the image of q. Here s agrees with the arclength coordinate on $w = 0$; it is not being identified with the first parameter of q^{-1} away from the core. Taylor's formula with integral remainder gives

$$h(s, w) = w^2 b(s, w)$$

with b smooth. Take a fixed smooth cutoff χ equal to one on $[-1, 1]$ and zero outside $(-2, 2)$. For small $\eta > 0$ put

$$h_\eta(s, w) = \chi(w/\eta) w^2 b(s, w), \qquad X_\eta = X_0 + h_\eta n$$

on the collar and $X_\eta = X_0$ elsewhere. For η small the support lies compactly in U. All the following estimates are uniform on the collar, with constants depending on the already fixed strip:

$$h_\eta = O(\eta^2), \quad (h_\eta)_s = O(\eta^2), \quad (h_\eta)_w = O(\eta),$$

$$(h_\eta)_{ss} = O(\eta^2), \quad (h_\eta)_{sw} = O(\eta), \quad (h_\eta)_{ww} = O(1).$$

Thus $X_\eta \to X_0$ in C^1, giving immersion and global embedding for small η. Its consistently chosen unit normal satisfies $n_\eta - n = O(\eta)$.

Let L_η, M_η, N_η be the second fundamental coefficients of X_η in the fixed collar. For the original surface $L_0(s, 0) = 0$ by asymptoticity, so $L_0(s, w) = O(w)$. Also $M_0(s, 0)$ is nowhere zero, because at the core the determinant of the second fundamental form is $-M_0(s, 0)^2 < 0$. The derivative estimates just stated and $n_\eta - n = O(\eta)$ imply on $|w| \le 2\eta$

$$L_\eta = O(\eta), \qquad M_\eta = M_0(s, 0) + O(\eta), \qquad N_\eta = O(1).$$

For example the ss derivative of the added normal displacement is $O(\eta^2)$, and the change of normal in the bounded original ss coefficient contributes $O(\eta)$; the mixed and ww estimates follow in the same way. Hence

$$L_\eta N_\eta - M_\eta^2 = -M_0(s, 0)^2 + O(\eta) < 0$$

uniformly for small η, since $\min_s |M_0(s, 0)| > 0$. The first fundamental determinant stays bounded below by a positive constant by C^1 convergence. Formula (1) proves $K_{X_\eta} < 0$ on the modified collar. Outside it the surface is unchanged, so the curvature is negative on all of U.

Fix η small enough for all the preceding conclusions and for $\|X_\eta - X_0\|_{C^1} < \varepsilon$. On $|w| < \eta$, $X_\eta = X_0 + hn = R \circ q^{-1}$ exactly. By compactness of the core, there is $a > 0$ such that the image under q of $|u| < a$ lies in $|w| < \eta$. Restricting q to this strip gives exactly the inclusion asserted in the statement. If X_0 is tight, Lemma 2.2 applies. □

Lemma 8.2 (An actual supported band gives a global bending)**.** *Let $X : M \to \mathbb{R}^3$ be an embedding of a closed surface, let $\ell > 0$, let $I \subset \mathbb{R}$ be a nonempty open interval, and put $A = (\mathbb{R}/\ell\mathbb{Z}) \times I$. Let $\iota : A \to M$ be a smooth annular embedding with $X \circ \iota = B$. If B has an infinitesimal bending Y_A compactly supported in A and with $\int_A \det(Y_A, B_s, B_c)\, ds\, dc \neq 0$, then X has an infinitesimal bending Y supported in $\iota(\operatorname{supp} Y_A)$ and with nonzero outward normal flux.*

Proof. An embedding of equal-dimensional manifolds has open image, by the inverse function theorem. Define $Y \circ \iota = Y_A$ on that image and $Y = 0$ outside it. The image of $\operatorname{supp} Y_A$ is a compact subset of the open set $\iota(A)$. Every point outside $\iota(A)$ therefore has a neighborhood disjoint from this image, on which the extended field is zero. Thus the extension is smooth. The chain rule and the literal identity $X \circ \iota = B$ transfer (2) to $\iota(A)$; outside the support it is zero. On this connected annulus the outward normal of X is either the parameter normal $(B_s \times B_c)/|B_s \times B_c|$ or its negative, with a single constant sign. The change-of-variables formula therefore gives

$$\int_M \langle Y, \nu_X \rangle \, dA_X = \pm \int_A \det(Y_A, B_s, B_c)\, ds\, dc \neq 0.$$

□

Proof of Theorem 1.6. Choose a collar with compact closure in U. Proposition 4.4 supplies an inflection on the original curve. Apply Proposition 5.1, with C^2 error as small as needed, supported in that collar. Call the resulting tight embedding X_1. Parametrize its actual new core by its own arclength and equip it with its Darboux frame. It satisfies precisely the hypotheses of Proposition 6.1.

Fix the ruling supplied by that proposition. Apply Proposition 8.1 to X_1 and this ruled strip, again with support compactly in U, to obtain X. Choose the errors in the two modifications so that their sum in C^1 is less than the prescribed ε. The surface remains negatively curved on all of U and is tight. The inclusion of a neighborhood of the ruled core is exact.

Let q and $a > 0$ be supplied by Proposition 8.1. Only now choose a bounded nonempty open interval I of sufficiently large leaf parameters c so that $c + \omega(s) > 0$ and $0 < u(s, c) < a$ for every $(s, c) \in (\mathbb{R}/\ell\mathbb{Z}) \times I$. The annulus $B(s, c) = \gamma(s) + \rho(s)V(s)/(c + \omega(s))$ lies inside that exact neighborhood. Set $\iota(s, c) = q(s, u(s, c))$. Then ι is an annular embedding and $X \circ \iota = B$. Theorem 7.1 supplies a field on this same annulus with compact support and nonzero signed flux. Lemma 8.2 produces the desired global field Y. Its compact support lies in U.

This proves every assertion in (4). The choices were made in the order dictated by their dependencies: the local preparation, then the ruling, then the insertion width, then the leaf interval, and finally the bending function. No limiting assertion about an earlier choice is used after a later choice has been fixed. □

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